The Bombay Salesian Society's

Don Bosco Institute of Technology, Kurla(W), Mumbai

Digital VLSI Design
Practice Questions

1. Write equation of drain current for NMOS transistor for cut-off region, linear region, and saturation region along with conditions.
2. Write equation of drain current for PMOS transistor for cut-off region, linear region, and saturation region along with conditions.
3. List various short channel effects in short channel MOSFET
4. List various process used in the fabrication of MOSFET
5. Tabulate all parameters of MOSFETs before CVS and after CVS
6. Tabulate all parameters of MOSFETs before CFS and after CFS
7. List various masks used in the fabrication process of NMOS transistor in the order of their use.
8. Numerical based of MOSFET current equations.
9. Numerical based on MOSFET scaling equations.
10. Draw cross section of MOSFET after processing of Metallization Mask along with metallization mask
11. Draw cross section of MOSFET after processing of Gate Mask along with metallization mask
12. Define long channel MOSFET and short channel MOSFET
13. What is difference between wet oxidation and dry oxidation? Where these processes are used during MOSFET fabrication.
14. Draw cross sections of all masks used in NMOS fabrication.
15. Explain hot carrier effect.
16. Explain Drain Induced barrier Lowering effect
17. What is velocity saturation and what is its impact of MOSFET performance
18. Draw VTC of CMOS inverter and locate all critical points on the same.
19. Define V_{IL}, V_{IH}, V_{OH}, V_{OL} and V_{INV} w.r.t VTC of CMOS Inverter
20. In CMOS inverter VDD=3.3V, Kn=150uA/V^2 and Kp=75uA/V^2, V_{TN}=0.7V and V_{TP}=-0.7V Find V_{OH}, V_{OL}, V_{INV}
21. Differentiate between Constant Voltage Scaling and Constant field Scaling with respect to I_{DSlin}, I_{DSsat}, Power, Area, Delay, Power Density.
22. Give fabrication flow of NMOS transistor with appropriate cross-sections and masks
23. Define constant voltage scaling and explain its impact on MOSFET parameters by deriving necessary equations
24. Define constant field scaling and explain its impact on MOSFET parameters by deriving necessary equations
25. By deriving expressions prove that delay of MOSFET reduces by factor of S^2 (S square) and by S in case of Constant Voltage Scaling and Constant Field Scaling respectively. S is scaling factor
26. Give fabrication flow of PMOS transistor with appropriate cross-sections and masks
27. Draw circuit diagram of CMOS inverter and VTC and explain its working
28. Draw circuit diagram and VTC of CMOS inverter and derive expression for V_{IH} and V_{IL}
29. Draw layout of CMOS inverter using lambda rules

30. Draw two input NAND gate and its equivalent inverter if (W/L) ratio of all transistors in NAND gates is 20
31. Design CMOS Inverter by calculating widths (W) NMOS and PMOS transistor to provide t_{PHL} =20ns and t_{pLH}=20ns. Use V_{DD}=5V, Kn=100uA/V^2 and Kp=50uA/V^2, V_{TN}=1V and VTP=-1.0V
32. Draw two input NOR gate and its equivalent inverter if (W/L) ratio of all transistors in NAND gates is 20
33. Draw layout of NAND gate using lambda rules
34. Draw layout of NOR gate using lambda rules
35. Numerical based on finding of Noise Margins, propagation delay etc
36. Implement given logic function using Standard CMOS, PTL, Transmission Gate, Pseudo NMOS
37. What are limitations of Pass Transistor logic and how to overcome the same.
38. Draw layout of complex circuit
39. Explain working of MOSFET in linear region, saturation region, cut off region using output characteristics, Also derive expression of Drain current in Linear region and Saturation region.
40. What are the various sources of power dissipation in CMOS inverter? Explain each source in detail